



Course: SWE-Computer Network Practical							
Instructor	Dr. Asad Raza Malik			Practical/Lab No.	13		
Date				CLOs	03		
Student's Roll no:				Point Scored:			
Date of Conduct:				Teacher's Signature:			
LAB PERFORMANCE INDICATOR	Subject Knowledge	Data Analysis and Interpretation	Ability to Conduct Experiment	Presentation	Calculation and Coding	Observation/Result	Score

Topic	To configure the EIGRP dynamic routing protocol on two routers
Objective	To configure the EIGRP dynamic routing protocol on two routers.

Lab Discussion: Theoretical concepts and Procedural steps

EQUIPMENT

- Two PC
- Two Routers with console
- Two RJ-45 TO DB-9 adapter
- Two RJ-45 TO RJ 45 rollover cable

TOOL

- PACKET TRACER DISCUSSION & CONFIGURATION

What is EIGRP?

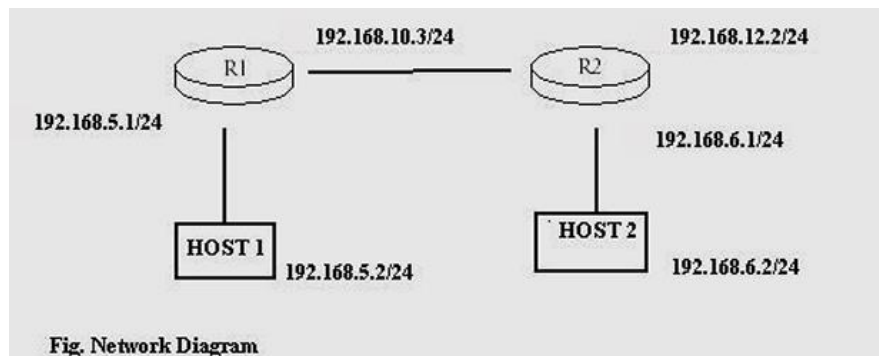
The Enhanced Interior Gateway Routing Protocol (EIGRP) represents an evolution from its predecessor IGRP ("Interior Gateway Routing Protocol"). This evolution resulted from changes in networking and the demands of diverse, large-scale internet works. EIGRP integrates the capabilities of link-state protocols into distance vector protocols. Additionally, EIGRP contains several important protocols that greatly increase its operational efficiency relative to other routing protocols. One of these protocols is the Diffusing update algorithm (DUAL). DUAL enables EIGRP routers to determine whether a path advertised by a neighbor is looped or loop-free, and allows a router running EIGRP to find alternate paths without waiting on updates from other routers.

EIGRP provides compatibility and seamless interoperation with IGRP routers. An automatic-redistribution mechanism allows IGRP routes to be imported into EIGRP, and vice versa, so it is possible to add EIGRP gradually into an existing IGRP network. Because the metrics for both protocols are directly translatable, they are as easily comparable as if they were routes that originated in their own autonomous systems (ASs). In addition, EIGRP treats IGRP routes as external routes and provides a way for the network administrator to customize them.

Key capabilities that distinguish EIGRP from other routing protocols include fast convergence, support for variable-length subnet mask, support for partial updates, and support for multiple network layer protocols.

A router running EIGRP stores all its neighbors' routing tables so that it can quickly adapt to alternate routes. If no appropriate route exists, EIGRP queries its neighbors to discover an alternate route. These queries propagate until an alternate route is found. Its support for variable-length subnet masks permits routes to be automatically summarized on a network number boundary. In addition, EIGRP can be configured to summarize on any bit boundary at any interface.

EIGRP does not make periodic updates. Instead, it sends partial updates only when the metric for a route change. Propagation of partial updates is automatically bounded so that only those routers that need the information are updated. As a result of these two capabilities, EIGRP consumes significantly less bandwidth than IGRP.



Setup a network similar to the one in the diagram. Any router that meets the interface requirements may be used. And follow the steps required to achieve this lab activity.

Step 1: Configuring both Routers

For Router1

Press Enter to Start

```
Router> Router>en
```

```
Router# configuration terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#hostname Router1 Router1
```

```
(config)#int s0
```

```
Router1(config-if)#ip address 192.168.10.3 255.255.255.0
```

```
Router1(config-if)#no shut
```

```
%LINK-3-UPDOWN: Interface Serial0, changed state to up
```

```
%LINK-3-UPDOWN: Interface Serial0, changed state to down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state to down
```

```
Router1(config-if)#exit
```

```
Router1(config)#int e0
Router1(config-if)#ip address 192.168.5.1 255.255.255.0
Router1(config-if)#no shut
%LINK-3-UPDOWN: Interface Ethernet0, changed state to up
Router1(config-if)#^Z %SYS-5-CONFIG_I: Configured from console by console
```

```
For Router2
Press Enter to Start
Router> Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router2
Router2(config)#int s0
Router2(config-if)#ip address 192.168.12.2 255.255.255.0
Router2(config-if)#clock rate 64000
Router2(config-if)#no shut
%LINK-3-UPDOWN: Interface Serial0, changed state to up
Router2(config-if)#exit
```

```
Router2(config)#int e0
Router2(config-if)#ip address 192.168.6.1 255.255.255.0
Router2(config-if)#no shut
%LINK-3-UPDOWN: Interface Ethernet0, changed state to up
Router2(config-if)#^Z
%SYS-5-CONFIG_I: Configured from console by console
```

a. How would you assign class A ip address to interface serial0 to Router2? Write here.

Step 2: Configuring the workstations

Configure the workstations1 with the proper IP address, subnet mask, and default gateway.

a. The configuration for the host connected to the Router1 is:

```
IP Address: 192.168.5.2
IP subnet mask: 255.255.255.0
Default gateway: 192.168.5.1
```

Configure the workstations2 with the proper IP address, subnet mask, and default gateway.

a. The configuration for the host connected to the Router2 is:

```
IP Address: 192.168.6.2
IP subnet mask: 255.255.255.0
Default gateway: 192.168.6.1
```

Step 3: Check the interface status

Router1#sh ip int brief

Interface	IP-Address	Ok? Method	Status	Protocol
Serial 0	192.168.10.3	Yes unset	up	up
Ethernet 0	192.168.5.1	Yes unset	up	up

Router2#sh ip int brief

Interface	IP-Address	Ok? Method	Status	Protocol
Serial 0	192.168.12.2	Yes unset	up	up
Ethernet 0	192.168.6.1	Yes unset	up	up

a. Check the interfaces on both routers with the commands show ip interface brief or show interface.

```
Router1#show interface
FastEthernet0/0 is administratively down, line protocol is down
(disabled)
Hardware is Lance, address is 000b.be88.2684 (bia 000b.be88.2684)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Full-duplex, 100Mb/s, media type is RJ45
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 input packets with dribble condition detected
  0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
--More-- |
```

b. Are all the necessary interfaces up?

Step 4: Check the routing table entries

Router1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i-
IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate U
default- per-user static route
Gateway of last resort is not set 192.168.5.0/24 is subnetted, 1 subnets
C 192.168.5.0 is directly connected, Ethernet0
192.168.10.0/24 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0

Router2#sh ip route

i - U Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D- EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2,
* - candidate default U - per-user static route
Gateway of last resort is not set 192.168.6.0/24 is subnetted, 1 subnets
C 192.168.6.0 is directly connected, Ethernet0 192.168.12.0/24 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0

a. What networks are displayed?

b. What interface is directly connected?

Step 5: Configuring the routing protocol on all the Routers

At the global configuration mode, enter the following:

```
Router1(config)#router eigrp 20
Router1(config-router)#network 192.168.10.0
Router1(config-router)#network 192.168.5.0
Router1(config-router)#^Z
```

```
Router2(config)#router eigrp 20
Router2(config-router)#network 192.168.12.0
Router2(config-router)#network 192.168.6.0
Router2(config-router)#^Z
```

Step 6: Verify the new route

Router1#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default U - per-user static route
Gateway of last resort is not set 192.168.5.0/24 is subnetted, 1 subnets
C 192.168.5.0 is directly connected,
Ethernet0 192.168.10.0/24 is subnetted, 1 subnets
C 192.168.10.0 is directly connected, Serial0 192.168.6.0/24 is subnetted, 1 subnets i
D 192.168.6.0 [90/1628160] via 192.168.12.2, 00:00:03, Serial0

Router2#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP i IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default- per-user static route

Gateway of last resort is not set 192.168.6.0/24 is subnetted, 1 subnets
C 192.168.6.0 is directly connected, Ethernet0 192.168.12.0/24 is subnetted, 1 subnets
C 192.168.12.0 is directly connected, Serial0 192.168.5.0/24 is subnetted, 1 subnets
D 192.168.5.0 [90/1628160] via 192.168.10.3, 00:00:57, Serial0

a. What are the entries in the ROUTER1 routing table?

b. What are the entries in the ROUTER2 routing table?

c. What is the administrative distance?

Step 7: Check connectivity from host to host

Ping host1 to host2

C:>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=60ms TTL=241

Reply from 192.168.6.2: bytes=32 time=60ms TTL=241

Reply from 192.168.6.2: bytes=32 time=60ms TTL=241

Reply from 192.168.6.2: bytes=32 time=60ms TTL=241

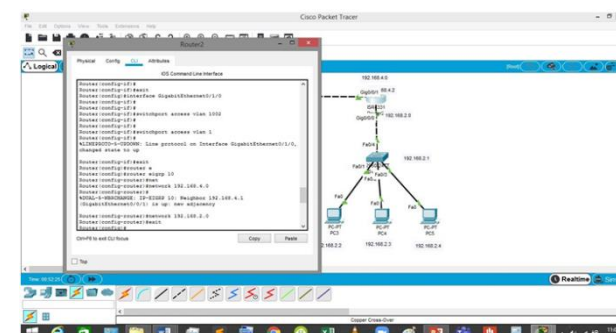
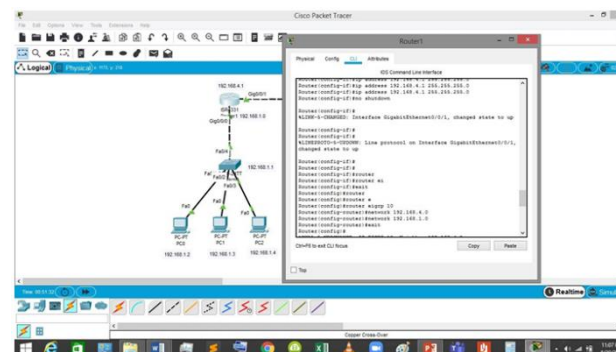
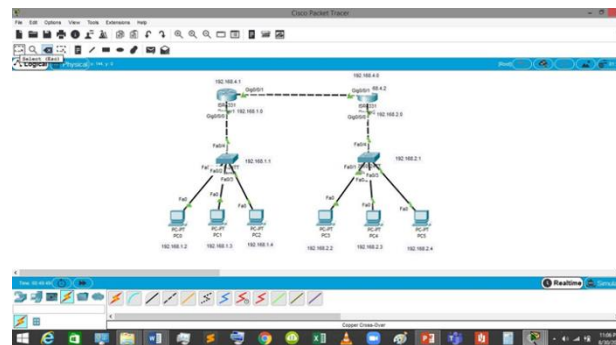
Reply from 192.168.6.2: bytes=32 time=60ms TTL=241

Ping statistics for 192.168.6.2:

Packets: Sent = 5, Received = 5, Lost = 0 (0% loss), Approximate round-trip times in milliseconds:

Minimum = 50ms, Maximum = 60ms, Average = 55ms

If the ping was not successful, check routing table to make sure static routes are entered correctly.



Tasks:

1. Perform the above task and write the answers according to your experiments.
2. Configure EIGRP protocol on 3 routers containing 3 computers each.
3. Configure EIGRP protocol on 5 routers containing 3 computers each.

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